MTIA v2: Meta's Second Generation AI Chip: Model-Chip Co-Design and Productionization Experiences

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Based on a paper published in ISCA 2025

MTIA 2i is optimized for inference

MTIA 2i reduces the TCO by ~44% over GPUs.

It can support Meta's latest models.

MTIA 2i-based production server, equipped with 24 MTIA 2i chips, achieves total performance comparable to that of our GPU-based production server equipped with eight GPUs.

Model Type	Description	Model Size	Model Complexity
Retrieval	Rank initial 1M candidates.	50-100 GB	0.001 - 0.01 GFLOPS/sample
Early stage ranking	Rank 10K candidates.	100-300 GB	0.01 - 0.1 GFLOPS/sample
Late stage ranking	Final ranking of top 100 candidates.	100-300 GB	0.2 - 2 GFLOPS/sample
HSTU retrieval	Rank initial hundreds of Ms candidates.	1TB	10 GFLOPS/request
HSTU ranking	Similar to ranking above.	2TB	80 GFLOPS/request

Table 1: Examples of some production models at Meta. Note that 90% of model size is embeddings.

		MTIA 2i	MTIA 1
Technology		TSMC 5nm	TSMC 7nm
Frequency		1.35GHz	800MHz
Instances		2.35B gates, 103M flops	1.12B gates, 65M flops
Area		25.6mm x 16.4mm	19.3mm x 19.1mm
Package		50mm x 40mm	43mm x 43mm
Voltage		0.85V	Dual rail: 0.67V (logic), 0.75V (memories)
TDP		85W (65W typical)	35W (25W typical)
Host connection		8x PCIe Gen5 (32 GB/s)	8x PCIe Gen4 (16 GB/s)
GEMM TOPS		354 TFLOPS/s (INT8)	102.4 TFLOPS/s (INT8)
		177 TFLOPS/s (FP16/BF16)	51.2 TFLOPS/s (FP16)
GEMM TOPS (Sparsity)		708 TFLOPS/s (INT8)	N/A
		354 TFLOPS/s (FP16/BF16)	N/A
SIMD TOPS	RISC-V vector core	5.5 (INT8), 2.8 (FP16), 1.4 (BF16/FP32)	3.2 (INT8), 1.6 (FP16), 0.8 (FP32)
	SIMD Engine	5.5 (INT8/FP16/BF16/FP32)	3.2 (INT8), 1.6 (FP16)
Memory capacity	Per-PE local memory	384 KB	128 KB
	On-chip SRAM shared across PEs	256 MB	128 MB
	Off-chip LPDDR5	64-128 GB	32-64 GB
Memory bandwidth	Per-PE local memory	1 TB/s	0.4 TB/s
	On-chip SRAM	2.7 TB/s	0.8 TB/s
	Off-chip LPDDR5	204.8 GB/s	176 GB/s

Compared to MTIA 1, MTIA 2i delivers >3x peak FLOPS, >3x SRAM bandwidth, >3x network-on-chip bandwidth, 2x DRAM capacity, and 1.4x DRAM bandwidth. Area is 1.13X.

LPDDR5
DDR Ctrl
MEM
Decompression
Control Core
XBAR
PE
Host Interface

For vector computation, MTIA 2i can utilize either the SIMD Engine or the RISC-V vector extension with 64B vector registers.

SIMD Engine offers 2x and 4x higher throughput for FP16 and BF16/FP32, respectively.

New features in MTIA v2

Reduction engine performs dynamic INT8 quantization by identifying minimum and maximum values per batch and then computing the scaling factors.

GZIP compression is done for PCIe communication between the host and device, alleviating PCIe and network congestion.

MTIA 2i supports 2:4 weight sparsity in the Dot Product Engine, which could potentially double effective FLOPS.

To better support PyTorch's eager mode, MTIA 2i provides hardware features that accelerate job launches.

It can launch jobs in 1us.

Eager mode executes operations immediately as they are called, rather than first compiling them into a static computation graph.

Each MTIA 2i server has two CPU sockets, each connected to a PCIe switch that links to six accelerator modules. Each module houses two MTIA 2i chips, connected via two 8x Gen5 PCIe links.

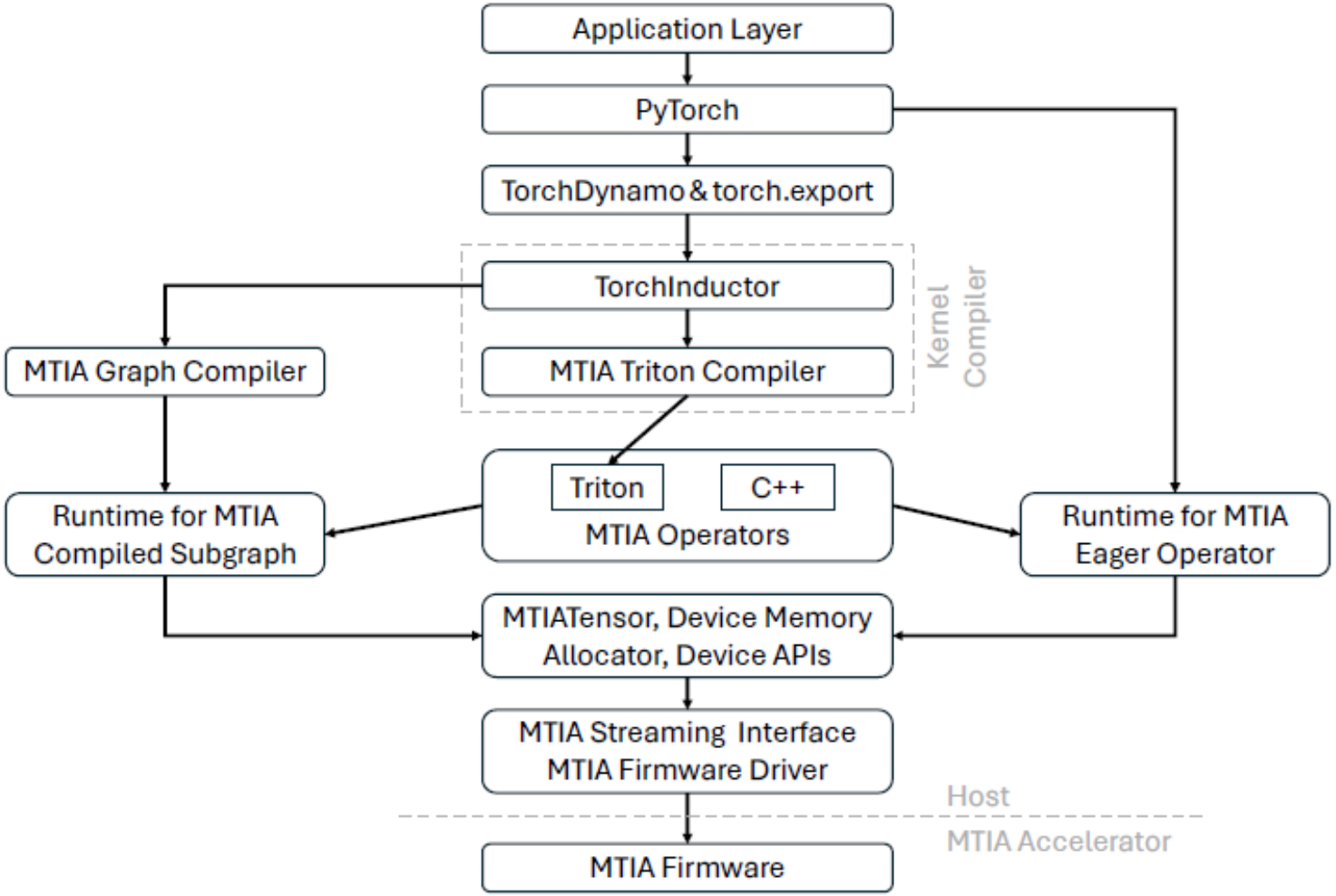
In total, each server contains two CPUs and 24 MTIA 2i accelerators.

Each CPU has 96 cores and is connected to 12 x 96GB DDR5 DRAM (totaling 1.15 TB per CPU), offering 460 GB/s of bandwidth.

Additionally, each CPU is equipped with 2 x 200Gbps Ethernet NICs.

==> Each accelerator has access to 8 CPU cores, 96 GB of host DRAM with 38 GB/s bandwidth, and 4.17 GB/s Ethernet bandwidth.

From MTIA 1 to MTIA 2i, the software stack has evolved from PyTorch with the FX compiler to PyTorch 2.0 [6] with TorchDynamo, TorchInductor, and Triton



MTIA 2i's SRAM provides 2.7 TB/s of bandwidth, whereas LPDDR offers just 204 GB/s—a 13x difference.

Compared to MTIA 1, we doubled the SRAM capacity and tripled its bandwidth, DRAM bandwidth is only 1.4x.

We also doubled the bandwidth of Local Memory to the Dot Product Engine (DPE) compared to MTIA 1.

The 256MB global SRAM shared across PEs is partitioned, at a granularity of 32MB, regions into hardware-managed cache (LLC) and software-managed scratch memory (LLS),

Using LPDDR instead of HBM reduces costs but introduces limitations that make MTIA 2i unsuitable for certain models.

With limited off-chip bandwidth, performance drops sharply as models reach a complexity and size that exceeds the SRAM capacity.

If each sample requires 2 GFlops/sample, then, GEMMs become DRAM bandwidth-bound. Then, latency of sparse and dense networks become too large. In fact, limited LPDDR bandwidth makes it challenging to run LLMs on MTIA 2i.

It is unsuitable for running large models such as Llama3 70B or 405B.

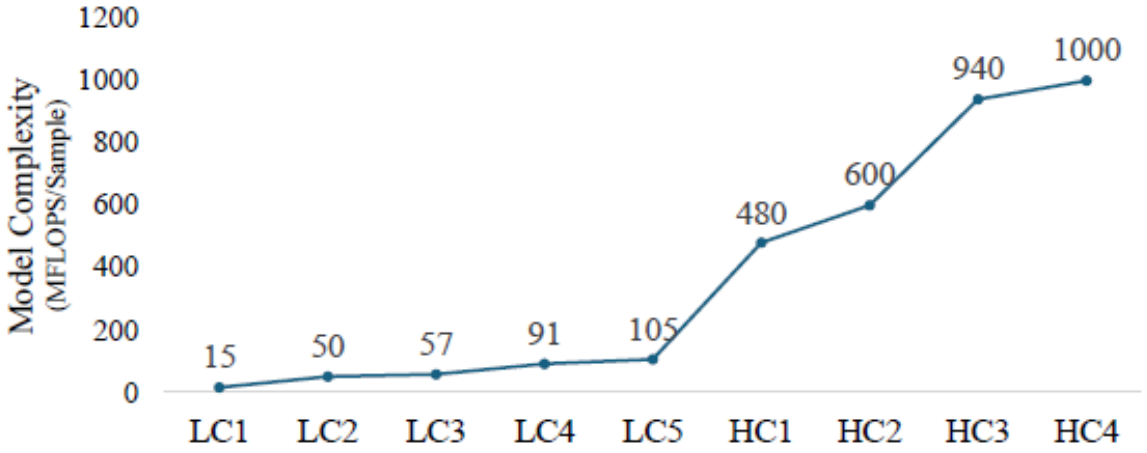
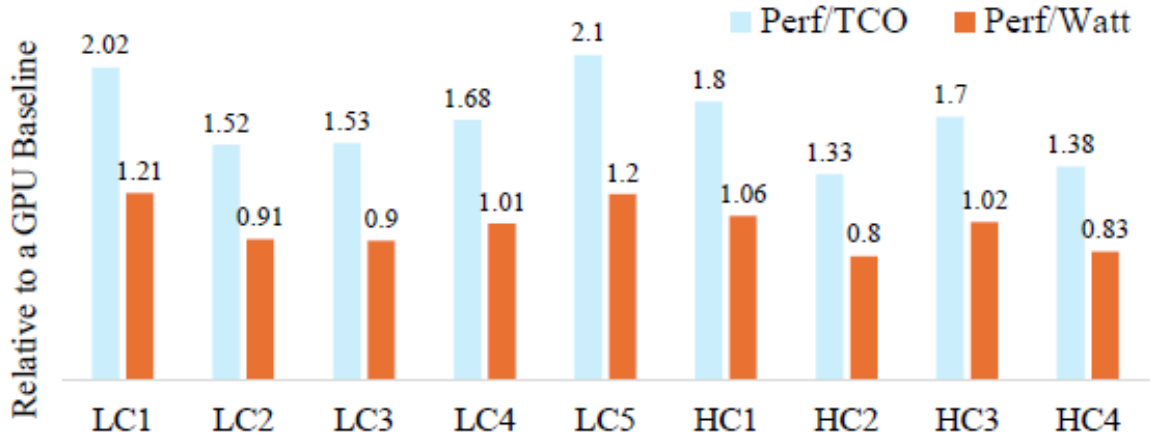
On increasing chip's operating frequency from 1.1GHz to 1.35GHz, they observed end-to-end throughput improvements ranging between 5% and 20%.

LPDDR DRAM, which lacks built-in support for Error-Correcting Code (ECC).

They found that enabling ECC is necessary for production-scale operation, despite the 10-15% throughput penalty associated with the inefficient memory-controller-based ECC.

Results

There is no clear trend for efficiency as a function of complexity, but the highest efficiency was achieved on LC models, while the lowest efficiency was observed on HC models.

This is consistent with the fact that a model's complexity and size determine how well it fits into the global SRAM and, therefore, how well it can utilize the Dot Product Engine and SIMD Engine.

When working with an in-house custom AI chip, it is easier to outperform GPUs in Perf/TCO than in Perf/Watt, due to GPUs having a much longer history in power optimization.